A-A-53327A
12 July 1991
SUPERSEDING
A-A-53327
26 February 1988

COMMERCIAL ITEM DESCRIPTION

CASSETTES, RADIOGRAPHIC FILM

(Heavy Duty Type)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for GG-C-131C which is cancelled.

This Commercial Item Description covers a heavy duty type radiographic film cassette with two high speed intensifying screens.

Salient Characteristics:

The radiographic film cassette shall be a heavy duty type radiographic film cassette consisting of a frame, a front, a back, a cushion, and a pair of high speed intensifying screens. The cassettes shall be of the following types:

Type I - Conventional; with high speed intensifying screens

Type II - Grid front; with high speed intensifying screens

The cassettes shall be of the following sizes:

Size 1 - 8 by 10 inches
Size 2 - 10 by 12 inches
Size 3 - 11 by 14 inches
Size 4 - 14 by 17 inches

The cassette frame shall be durable, rigid, and girder type, constructed of corrosion and warp resistant metal with reinforced or welded corners. The frame shall be flat within 1/32 inch.

AMSC N/A FSC 6525

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The cassette front panel shall be constructed of a suitable magnesium or aluminum material fitted to the frame so as to preclude buckling or warping The panel shall be covered with a suitable plastic or resin coating and shall be flat within 1/32 inch, and its x-ray opacity shall be uniform and free from signs of foreign matter. A melamine type finish on the plastic facing shall be acceptable.

The cassette back shall be constructed of a suitable lightweight, durable metal, lined with approximately 0.005 inch 0.13 of lead equivalent measured with radiation of a potential difference of 100 KV. The back shall be linked to the frame on the short dimension by self-centering, free-floating, easily operable hinges of suitable stainless steel. The hinge shall permit the door to open a minimum of 180° to allow easier removal of the film after an exposure has been made.

A suitable stainless steel D ring shall be securely attached to the outside of the back, opposite the hinged end, to facilitate opening the cassette. The back shall be designed to lift the film when opened. All screws shall be corrosion-resisting steel or, as an alternate, carbon or alloy steel, cadmium plated to resist corrosion.

The cassette back shall have at least two properly spaced crossbars of spring-tempered stainless steel securely riveted in the center to the back of the cassette The crossbar locks shall have a minimum No. 4 commercial finish. The crossbars shall be designed so that when the cassette is unloaded they are rotatable 360° When the cassette is loaded and the crossbars are in the locked position, they shall, in conjunction with pressure cams properly formed into the cassette back and a resilient cushion, provide evenly distributed pressure for uniform screen/film contact.

The cassette cushions shall be securely mounted on the inside of the cassette back and on the inside of the front panel of the cassette and shall be of such size and material that it will prevent any light from leaking to the radiographic film. The adhesive used for mounting the cushion shall be of a material that will not adversely affect the cushion or intensifying screens, and will not leave shadows on the radiographic film exposed within the cassette

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The type II cassettes only, shall be furnished with a grid. The grid shall be the linear focused type having the following characteristics:

Ratio	6:1
Lines per inch	90
Focal range	28 to 72 inches
Thickness	0.1 inch
Weight, per square inch	0.14 ounce

The grid shall consist of narrow lead strips separated by fiber or plastic interspaces, transparent to x-rays. The entire grid shall be sealed in a strong, durable, low density aluminum envelope impervious to moisture. The grid shall be fitted snugly into the cassette frame, and shall be retained in position by the frame stiffeners. The grid shall be removable by removing the frame stiffeners. There shall be no fasteners in the grid area.

The intensifying screens shall be of a high speed type with a speed factor of not less than 1.6. The screens shall be mounted on a thin sheet of polyurethane foam on the cassette front and back to optimize film-screen contact. The screens shall be in accordance with the latest revision of ANSI Standard PH1.50, Dimensions for Radiographic Film Cassette Intensifying Screens. This commercial item description shall govern in the event it deviates from the ANSI Standard

The cassette shall be in accordance with the latest revision of ANSI Standard PH1 49, Specifications for Medical Radiographic Film Cassettes, except the thickness of the cassette shall not be less than 0.600 inch, nor more than 0.690 inch This commercial item description shall govern in event it deviates from the ANSI standard.

The cassette finish shall be free of chips, cracks, scratches, and other appearance defects.

The cassette shall be clean and free of dirt, oil, and grease.

Workmanship. The radiographic film cassette shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA) One cassette of the type and size specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 4.0 percent defective) and an inspection level of II The unit of product for examination purposes shall be one radiographic film cassette.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-3. The unit of product for test purposes shall be one radiographic film cassette.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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General test. Test radiographs shall be made at a focal spot to film distance of 40 plus or minus 1 inch, at 80 Kvp, with screen-type, medium speed x-ray film. The x-ray tube shall be perpendicular to the center of the cassette. The nominal focal spot size of the x-ray tube shall not be more than 1.0 mm. A minimum of 2.3 mm of aluminum filtration shall be in the useful beam. The exposed x-ray film shall be developed in accordance with the manufacturer's instructions. The developed x-ray film shall be viewed with an unaided eye using a fluorescent type x-ray film illuminator.

Drop test. The selected sample shall be dropped on each corner from a height of 30 inches onto a concrete floor. The cassette shall be dropped so that the plane of the cassette is perpendicular to the floor and the center of gravity of the cassette is directly over the corner which strikes the floor. Except for deformation at the impacted corner area, the cassette shall show no other sign of damage Any other kind of damage, shall constitute a defect After the drop test, the sample cassette shall be subjected to and pass the flatness test, lightproof test, and film contact test. For the type II cassette, the grid shall be removed from the cassette prior to the drop test.

Flatness test. The test for flatness shall be conducted on the open and closed cassette. The cassette frame shall rest on a flat test surface and any height variation shall be measured by passing feeler gages between the flat surface and the frame. The deviation from flatness of the front panel shall be determined by measuring the maximum distance between the front face and a straight edge placed at various angles on the front panel. The damaged corners resulting from the drop test shall not be placed on the flat test surface, and shall not be measured. Both frame and front panel shall be flat within 1/32 inch. Any deviation shall constitute a defect

X-Ray opacity of the front panel test. A non-screen type radiographic film shall be exposed in a cardboard, or plastic film holder, through the front panel with an exposure time to produce a diffuse transmission density of 1.0 plus or minus 0.2. The developed film shall be examined for uniformity of opacity and for evidence of included foreign matter. Visible shadows or non-uniformity shall be cause for rejection.

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Lightproof test. A high speed radiographic film, of the applicable size, shall be placed in the cassette, and all sides and edges of the cassette shall be exposed to a 100 watt incandescent lamp, operated at its rated voltage, at a distance of not more than one foot for not less than one minute. The developed test film shall be compared with a developed, unexposed film of the same lot. A density on the test film greater than that on unexposed film shall be cause for rejection When viewed against the illuminator, the test film shall show no density variations, except that fog within 1/8 inch of film border or within 1/4 inch square at the film corners shall not be considered a defect

Cassette liners test. A radiographic film in the sample cassette shall be exposed with an exposure time to produce a diffuse transmission density of 1.0 plus or minus 0.2. Shadows and artifacts on the test film shall be cause for rejection of the cassette.

Film contact test. Contact shall be determined by using a flat sheet of non-magnetic, bare metal, .039 plus or minus .004 inch thick, continuously perforated with holes of .098 plus 0, minus .02 inch diameter, at intervals of .16 plus 0, minus .02 inch. A flat screen of 1/4 inch mesh, made up of not more than 0 032 inch diameter, non-magnetic, bare, unsoldered wire may be used as an alternate. To facilitate the measurement of density of the film exposed during the test, the test object may have a clear area, 1/2 to 1 inch diameter, near its center. The test object shall be placed on the cassette using a cone which covers the film area but equally cuts the four film corners Exposure shall be made to produce a diffuse transmission density of 2 to 3. Uniformity of film contact shall be indicated by a uniformly sharp exposure through the mesh The test shall be repeated after exposure of the cassette to a minimum temperature of 120°F and a relative humidity of 80 plus or minus 2 percent for 48 hours The intensifying screens may be removed during the conditioning period

Grid test. The grid ratio and lines per inch shall be determined by measurement from radiographs. Radiographs of the grid shall be made as specified in the General paragraph at focal spot to film distances of 28 and 72 inches. The radiograph shall show that the space between lead strips is uniform in opacity and free from foreign matter. Non-uniform opacity or visible foreign matter shall constitute a failure

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document.

6525-00-226-0918 (Type II, Size 2)
6525-00-226-0919 (Type II, Size 4)
6525-00-721-9813 (Type I, Size 1)
6525-00-721-9814 (Type I, Size 2)
6525-00-721-9815 (Type I, Size 3)
6525-00-721-9816 (Type I, Size 4)

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA - OSS
USPHS
FDA-MPQAS

Project Number: 6525-1277

Location: CASSMC13.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-53327A	12 JULy 1991

3. DOCUMENT TITLE

CASSETTES, RADIOGRAPHIC FILM, (Heavy Duty Type)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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